

k -char p field

$$\sigma: k \rightarrow k$$

$$x \mapsto x^p$$

$$T = \text{Spec } B$$

$$\downarrow s$$

$$\text{Spec } k$$

Def. (1) Frobenius twist $T^{(p)} := T \times_{\text{Spec } k, \sigma} \text{Spec } k = \text{Spec } (B \otimes_{k, \sigma} k)$

$$\text{i.e. } \begin{array}{ccc} T^{(p)} & \longrightarrow & T \\ \downarrow & & \downarrow s \\ \text{Spec } k & \xrightarrow{\text{Spec } \sigma} & \text{Spec } k \end{array}$$

$$\begin{array}{ccc} k & \xrightarrow{\sigma} & k \\ \downarrow & & \downarrow \\ B & \longrightarrow & B \otimes_{k, \sigma} k \end{array}$$

(2) absolute Frobenius ; $\sigma: B \rightarrow B$
 $b \mapsto b^p \rightsquigarrow \text{Frob}_T: \text{Spec } B \rightarrow \text{Spec } B$

(for non-affine schemes, taking p^{th} power on sections defines a morphism of sheaves)

(3) relative Frobenius; call $T \dashrightarrow T^{(p)}$ the relative Frobenius φ_T .

$$\begin{array}{ccc} T & \xrightarrow{\text{Frob}_T} & T \\ \downarrow s & \searrow \exists! & \downarrow \\ T^{(p)} & \longrightarrow & T \\ \downarrow & & \downarrow s \\ \text{Spec } k & \xrightarrow{\text{Spec } \sigma} & \text{Spec } k \end{array}$$

(4) $r \geq 1$. By induction, p^r -Frobenius twist is $T^{(p^r)} := (T^{(p^{r-1})})^{(p)}$

relative p^r -Frobenius is $\varphi_T^r := \varphi_{T^{(p^{r-1})}} \circ \varphi_T^{r-1}$.

Concretely, $T^{(p^r)} \cong T \times_{\text{Spec } k, \sigma^r} \text{Spec } k = \text{Spec } (B \otimes_{k, \sigma^r} k) =: \text{Spec } (B^{(p^r)})$.

$$\sigma^r: B \rightarrow B$$

$$b \mapsto b^{p^r} \rightsquigarrow \text{Frob}_T^r: T \rightarrow T$$

$$\varphi_T^r = (\text{Frob}_T^r, s): T \rightarrow T^{(p^r)}$$

$$\uparrow$$

$$B^{(p^r)} \rightarrow B$$

$$b \otimes x \mapsto x \cdot b^{p^r}$$

$$\text{ex. } B = k[t_1, \dots, t_n] / (f_1, \dots, f_m)$$

$$\bullet B^{(p^r)} \cong k[t_1, \dots, t_n] / (f_1^{(p^r)}, \dots, f_m^{(p^r)}) \quad \text{where } f = \sum a_i t^i \rightsquigarrow f^{(p^r)} = \sum a_i^{p^r} t^i$$

$$\text{follows from } k[t_1, \dots, t_n] \rightarrow k[t_1, \dots, t_n] \otimes_{k, \sigma^r} k$$

$$\sum a_i t^i \mapsto \sum a_i t^i \otimes 1 = \sum t^i \otimes a_i^{p^r} \longleftrightarrow \sum a_i^{p^r} t^i \in k[t_1, \dots, t_n]$$

Formal consequences of the properties of fibre product: $\downarrow \text{Spec } k$.

$$\bullet (T_{\text{Spec } k}^x U)^{(p)} \cong T_{\text{Spec } k}^{(p)} \times_{\text{Spec } k} U^{(p)}$$

$$\bullet \varphi_{T_{\text{Spec } k}^x U} = (\varphi_T, \varphi_U).$$

$$\bullet \begin{array}{ccc} T & \xrightarrow{f} & U \\ \downarrow \text{Spec } k & \swarrow & \\ & & \end{array} \rightsquigarrow \begin{array}{ccc} T & \xrightarrow{\varphi_T} & T^{(p)} \\ \downarrow & & \downarrow \\ U & \longrightarrow & U^{(p)} \end{array} \quad \text{call } f^{(p)}: T^{(p)} \dashrightarrow U^{(p)} \text{ the Frobenius twist of } f.$$

$$\bullet (\varphi_T^r)^{(p^s)} = \varphi_{T^{(p^s)}}^r$$

in particular $\bullet G$ finite flat/ $k \Rightarrow G^{(p)}$ finite flat/ k .

$\bullet \varphi_G$ group scheme hom.

$$\bullet (G^{(p)})^\vee \cong (G^\vee)^{(p)}, \text{ hence } G^{(p)} \cong ((G^\vee)^{(p)})^\vee.$$

$\bullet f: G \rightarrow H$ group hom $\rightsquigarrow f^{(p)}: G^{(p)} \rightarrow H^{(p)}$ group hom. and

$$\begin{array}{ccc} G & \xrightarrow{\varphi_G} & G^{(p)} \\ f \downarrow & & \downarrow f^{(p)} \\ H & \xrightarrow{\varphi_H} & H^{(p)} \end{array}$$

$$\bullet f: G \hookrightarrow H \Rightarrow f^{(p)}: G^{(p)} \hookrightarrow H^{(p)}$$

$$\bullet f: G \xrightarrow{\sim} H \Rightarrow f^{(p)}: G^{(p)} \xrightarrow{\sim} H^{(p)}.$$

$$\text{Def. (Verschiebung)} \quad \varphi_{G^\vee}: G^\vee \rightarrow (G^\vee)^{(p)} \rightsquigarrow \psi_G: \begin{array}{ccc} (G^\vee)^{(p)} & \xrightarrow{\sim} & (G^\vee)^{\vee} \\ \parallel & & \parallel \\ G^{(p)} & \xrightarrow{\sim} & G \end{array}$$

$$\psi_G^r = \psi_G^{r-1} \circ \psi_{G^{(p)}}: G^{(p^r)} \rightarrow G.$$

Lemm. G, H finite flat/ k .

$$(a) \quad \psi_{G_{\text{Spec } k}^x H} = (\psi_G, \psi_H) \text{ via } (G_{\text{Spec } k}^x H)^{(p)} \cong G^{(p)} \times_{\text{Spec } k} H^{(p)}.$$

$$(b) \quad G \xrightarrow{f} H \rightsquigarrow f^{(p)}: G^{(p)} \rightarrow H^{(p)} \text{ st. } \begin{array}{ccc} G & \xleftarrow{\psi_G} & G^{(p)} \\ f \downarrow & & \downarrow f^{(p)} \\ H & \xleftarrow{\psi_H} & H^{(p)} \end{array}$$

Prop. (1) $\varphi_{\alpha_p} = 0 = \psi_{\alpha_p}$.

(2) $\varphi_{\mu_p} = 0$, $\psi_{\mu_p} = \text{id}_{\mu_p}$.

(3) $\varphi_{\mathbb{Z}/p\mathbb{Z}} = \text{id}_{\mathbb{Z}/p\mathbb{Z}}$, $\psi_{\mathbb{Z}/p\mathbb{Z}} = 0$.

Proof. (1) Recall $\alpha_p = \text{Spec } k[t]/(t^p)$, so φ_{α_p} maps t to $t^p = 0$.

Since $\alpha_p^\vee \cong \alpha_p$, it follows that $\psi_{\alpha_p} = 0$.

(2) similarly $\mu_p = \text{Spec } k[t]/(t^{p-1})$ so $\varphi_{\mu_p}(t) = t^p = 1$, which is the trivial homomorphism. ψ_{μ_p} will follow from (3), as $\mu_p^\vee = \mathbb{Z}/p\mathbb{Z}$.

(3) recall $\mathbb{Z}/p\mathbb{Z} = \text{Spec}(\prod_{\mathbb{Z}/p\mathbb{Z}} k)$. $\varphi_{\mathbb{Z}/p\mathbb{Z}}\left(\sum_{i \in \mathbb{Z}/p\mathbb{Z}} c_i e_i\right) = \sum_{i \in \mathbb{Z}/p\mathbb{Z}} c_i e_i^p = \sum_{i \in \mathbb{Z}/p\mathbb{Z}} c_i e_i$, as $e_i = (0, \dots, 1, \dots, 0)$ with 1 at i . So $\varphi_{\mathbb{Z}/p\mathbb{Z}} = \text{id}$, and $\psi_{\mathbb{Z}/p\mathbb{Z}}$ follows from (2). $\square$

Prop. $G = \text{Spec } A$ finite flat/ k . $S_n \curvearrowright A^{\otimes n}$ by permuting pure tensor factors.

(1) $\exists k$ -alg hom $\gamma: (A^{\otimes n})^{S_n} \rightarrow A^{(p)}$ st.

(i) $\gamma(a^{\otimes p}) = a \otimes 1$

(ii) for $v = a_1 \otimes \dots \otimes a_n$ with some $a_i = a_j$,

$$\gamma\left(\sum_{\sigma \in S_n / \text{stab}(v)} \sigma(v)\right) = 0.$$

(2) $\psi_G \rightsquigarrow \psi_A: A \rightarrow A^{(p)}$ fits into

$$\begin{array}{ccccc} & & \xrightarrow{\psi_A} & & \\ & \searrow & & \nearrow & \\ A & \longrightarrow & (A^{\otimes n})^{S_n} & \xrightarrow{\gamma} & A^{(p)} \\ & \searrow & \downarrow & & \\ & & A^{\otimes p} & & \end{array}$$

comultiplication of G .

Proof. as k -algebras, we have natural

$$A \cong (A^\vee)^\vee, \quad (\text{Sym}^p A^\vee)^\vee \cong (A^{\otimes p})^{S_n}, \quad A^{(p)} \cong ((A^\vee)^{(p)})^\vee.$$

also $k \text{ char } p \Rightarrow (f_1 + f_2)^{\otimes p} = f_1^{\otimes p} + f_2^{\otimes p} \in \text{Sym}^p A^\vee$. So

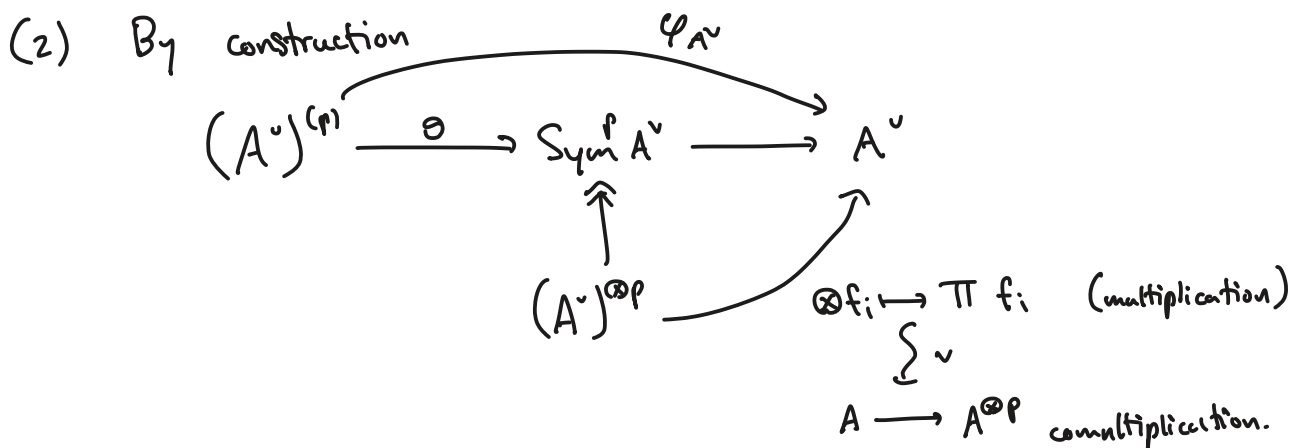
(1) $\exists! \theta: (A^\vee)^{(p)} \rightarrow \text{Sym}^p A^\vee$

$$f \otimes c \mapsto c \cdot f^{\otimes p}$$

$\gamma := \theta^\vee$. (i) for $a \in A \longleftrightarrow e_a \in (A^\vee)^\vee$ and $f \otimes c \in (A^\vee)^{(p)}$,

$$\gamma(a^{\otimes p})(f \otimes c) = e_a^{\otimes p}(c \cdot f^{\otimes p}) = c \cdot f(a)^{\otimes p} = f(a) \otimes c = (e_a \otimes 1)(f \otimes c).$$

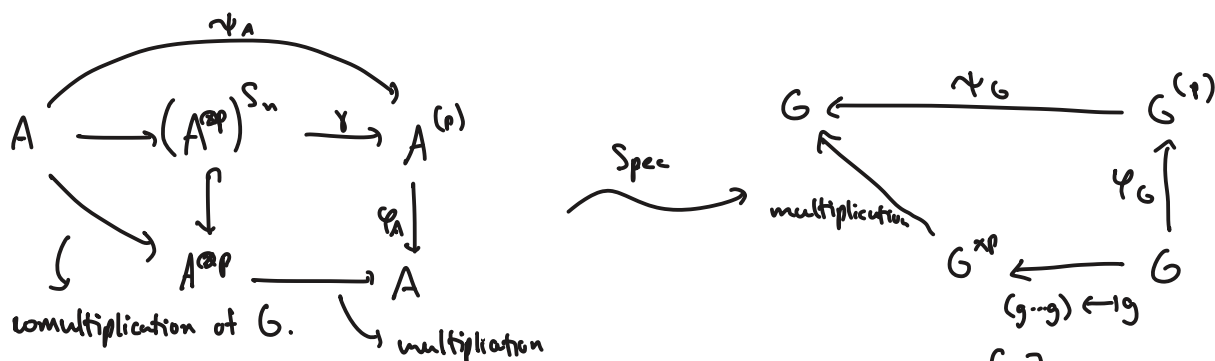
(ii) $\gamma\left(\sum_{\sigma \in S_n / \text{stab}(v)} \bigotimes_{i=1}^p a_{\sigma(i)}\right)(f \otimes c) = \sum_{\sigma} \left(\bigotimes_{i=1}^p e_{a_{\sigma(i)}}\right)(c \cdot f^{\otimes p}) = c \sum_{\sigma} \prod_{i=1}^p f(a_{\sigma(i)}),$ which is 0 as $p \nmid \sigma$



so the diagram dualizes to (2). $\square$

Prop. $G = \text{Spec } A$ finite flat/ k . Then $\psi_G \circ \varphi_G = [p]_G$ and $\varphi_G = \psi_G = [p]_{G^{(1)}}$.

Proof. The diagram from prev. prop extends to



So $\psi_G \circ \varphi_G = [p]_G$.

Moreover $\varphi_G^{(p)} = \varphi_{G^{(p)}}$ so

$$\begin{array}{ccc}
 G^{(p)} & \xrightarrow{\varphi_{G^{(p)}}} & G^{(p^2)} \\
 \downarrow \psi_G & & \downarrow \psi_{G^{(p)}} \\
 G & \xrightarrow{\varphi_G} & G^{(1)}
 \end{array}
 \Rightarrow \varphi_G \circ \psi_G = \psi_{G^{(1)}} \circ \varphi_{G^{(1)}} = [p]_{G^{(1)}}. \quad \square$$

Lem. $G = \text{Spec } A$ finite flat/ k . (1) φ_G iso $\iff$ injective

(2) G cunctd $\Rightarrow A$ artinian local k -alg whose max ideal is the augmentation ideal of G
 (recall: $I = \ker \epsilon$ counit.)

Proof. (1) recall from last time $\ker \varphi_G \hookrightarrow G$.

then Grothendieck's theorem $\Rightarrow 0 \rightarrow \ker \varphi_G \rightarrow G \rightarrow G/\ker \varphi_G \rightarrow 0$.

Now if $\ker \varphi_G = 0$ then $G \cong G/\ker \varphi_G$.

(2) also last time: G cunctd $\iff A$ finite local k -alg.
 $\xrightarrow{\text{alg}} \text{finite } k\text{-algebra is artinian.}$

$A/I \cong \text{im}(\epsilon)$ and $\text{im}(\epsilon) = k$ so I is the max idl. $\square$

Prop. (will not prove, appending past results)

(1) G cunctd $\iff \varphi_G^r$ vanishes, same r (can deduce for ψ criterion)

(2) G etale $\iff \varphi_G$ iso.

Prop. $G = \text{Spec } A$ finite flat/ k .

(1) $\ker(\varphi_G) = \text{Spec}(A/I^{(p)})$. $I^{(p)}$ is the ideal gen. by a^p , $a \in I$.

(2) if $\varphi_G = 0$ then $\dim G = p^{\dim_k I/I^2}$.

Proof.

Prop. G finite flat/ k , conctd. Then $\text{ord}(G) = p^r$. (some r)

Prop. $G = \text{Spec } A$ finite flat/ k , unit section e . Then $t_{G,e} \cong \text{Hom}_{k\text{-gp}}(G^\vee, G_a)$.

Prop. G ff/ k étale $\Leftrightarrow \text{Hom}_{k\text{-gp}}(G^\vee, G_e)$ vanishes.

Thm. $k = \bar{k}$. (1) simple ff k -gp is either étale or conctd.

(2) simple finite étale k -gps are $\mathbb{Z}/l\mathbb{Z}$, l prime.

(3) simple conctd ff k -gps are μ_p, α_p .